

ParaEMT EMT-validation campaign (IAS2026 / TX4)

Final report

Branch `research/parent-emt-validation` (from `a70dac94`; not pushed)

2026-09-11

Status: BLOCKED at the device gates. The network and operating point were reproduced (gate G1 PASS). The preregistered unit tests of the custom synchronous machine (G3) and the custom grid-following converter (G4) both FAILED. Under the preregistered stopping rules, EMT04–EMT18 were not run. No portfolio, policy, design, remediation or nonlinear claim receives EMT evidence from this campaign.

preregistration	<code>docs/20260911_PAREMT_EMT_PREREG_V1.md</code> , commit <code>c2947bd8</code> , before any TX4 EMT result; amendment A1 (<code>9d06436a</code>), before the unit-test results
simulator	NatLabRockies/ParaEMT_public, commit <code>d79d735a4a587d56c5b88187d1a499195b6b2b84</code> (2026-01-15), serial LU
environment	<code>.venv/xtool-parent</code> , CPython 3.11.10, exact upstream requirements

Erratum (V2-0 ledger reconciliation; `docs/20260911_PAREMT_EMT_LEDGER_RECONCILIATION.md`). The first version of this report took claim statuses from the pre-final ledger `results/20260911_CLAIM_MATRIX.csv` instead of the final canonical ledger `results/20260911_FINAL_VALIDATION_MATRIX.csv` (V01–V30). Corrected in place: (1) **I04/V21**. “I04 remains NOT YET TESTED” and “single-implementation results” were wrong. The custom-GFL results were independently reproduced in ANDES with the same equations (post-cumulant Phase 8, GATE 5 PASS; V21 VALIDATED as a reproduction of computation). EMT reproduction was not achieved. (2) **T08**. The Kundur holdout is 28/29 under the preregistered rule (the miss is a real-pair coalescence), i.e. 28/28 crossings through the origin. (3) **Section 23 and the closing table**. Regenerated from the final ledger. **Unchanged:** every EMT measurement, gate and verdict (G1 PASS, G3 FAIL, G4 FAIL, EMT04–EMT18 BLOCKED).

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1 Executive summary

Question. Do the principal observable predictions of the frozen phasor-domain theory survive an independent electromagnetic-transient (EMT) realization of the *same* IEEE-39 network and the *same* custom device and control equations?

Answer. This campaign cannot say. Each stage stopped at a preregistered gate:

1. **EMT00 (tool).** The official ParaEMT IEEE-39 case installs, runs deterministically, serializes and resumes.
 - Its no-event run is *not* stationary: a 0.62 pu start-up jump.
 - Cause: upstream never applies the one transformer ratio its case carries. With that tap applied, the jump falls to 2.2×10^{-4} pu.
 - The tool gate was recorded as “PASS (tool)”. The stationarity check as first written failed; this is disclosed in Section 4.
2. **EMT01 (network, G1): PASS.**
 - The frozen TX4 network is built from ParaEMT’s own companion elements.
 - Its 60-Hz admittance matches the canonical Ybus to 1.7×10^{-16} .
 - The EMT equilibrium matches the canonical power flow to 1.5×10^{-6} pu in $|V|$ and 5.9×10^{-6} rad in angle (tolerances 10^{-4} and 10^{-3}).
 - The official ParaEMT IEEE-39 differs from TX4 in 24 of 195 network rows, so it is not used.
3. **EMT02 (two-axis machine, G3): FAIL.**
 - Six of seven states track the phasor reference within 0.87% of their excursion, and the ringdown agrees ($\Delta\alpha = -3.1 \times 10^{-4} \text{ s}^{-1}$, $\Delta f = 1.2 \times 10^{-4} \text{ Hz}$).
 - The AVR state e_{fd} misses the 2% trajectory limit: 2.26% .
 - Diagnostics (not gates) attribute the excess to the electromagnetic dynamics of the EMT line, a physical effect of relative size $f/f_0 \approx 2\%$. The machine transcription itself agrees to 2×10^{-4} .
4. **EMT03 (11-state GFL, G4): FAIL.**
 - With the preregistered EMT line, an undamped step-to-step (Nyquist) mode of the trapezoidal line inductance is amplified by the ideal-current-source converter.

- At $g = 0$ the three runs go non-finite at 0.76s. At $g > 0$ they stay finite but saturate at a 0.11 pu alternation, giving state errors 3–1100 $\times$ the excursion.
- With the line realized algebraically (diagnostic only), all nine cases agree within 0.78 %.

5. EMT04–EMT18: BLOCKED.

What changes in the claim ledger. Nothing is weakened or strengthened, except that V19/I01 (network and operating point) gains an EMT reproduction. V21/I04 (the custom-GFL results in a second simulator) keeps its final-ledger status: **VALIDATED** by the independent ANDES reproduction (Phase 8). The EMT reproduction was not achieved. (*Corrected in V2-0; the first version wrongly said “NOT YET TESTED”.*)

Next step. A preregistration v2 is proposed for the author’s decision (Appendix A). It is not executed here.

2 Scientific questions

Q-EMT. Does an EMT realization of the same network and the same device equations reproduce the following observable consequences of the frozen phasor theory?

- the P4 portfolio trap (15 proper subsets of $\{30, 33, 35, 37\}$ stable, the full set unstable);
- the controller-only policy boundary ($g^* \approx 0.20768$) and the k boundary ($k^* \approx 1.3046$);
- port-guided design (the Newton direction; line reinforcement);
- the model-dependent remediations (condenser, governors);
- the near-Hopf finite-disturbance behaviour and the robustness holdout.

Precondition (gate A, MODEL). The network, the equilibrium and each custom device must first be shown to be realized equation-equivalently in EMT (EMT01–EMT04). This precondition failed at EMT02 and EMT03.

3 Evidence taxonomy

Every claim receives exactly one label:

- THEOREM (no EMT validation required);
- EMT CORROBORATED;
- EMT QUANTITATIVELY REPRODUCED;
- EMT REFUTED;
- EMT UNRESOLVED;
- NOT APPLICABLE TO EMT.

Wording. EMT can corroborate the *observable consequence* of a theorem. It never proves one.

- T01–T06 and C01–C06 are not EMT targets.
- Connected cumulants remain secondary and get no EMT figure.

In this report, “BLOCKED” means “not run because a preregistered stopping gate failed”. Its taxonomy label is EMT UNRESOLVED.

4 ParaEMT provenance and environment

Upstream.

- https://github.com/NatLabRockies/ParaEMT_public.git, commit d79d735a (“Update README.md”, 2026-01-15), cloned 2026-09-11.
- `external/ParaEMT_upstream` is pristine (empty diff).
- License: `LICENSE.md` (Alliance for Sustainable Energy, LLC).

Working copy. `external/ParaEMT_tx4`, same commit, changed only by `experiments/parent_empt/tx4_patch_parent.py`:

- `lib_numba.py` and `Lib_BW.py`, +86/–54 lines:
 - exact from-side transformer taps in `numba_InitNet`, `numba_updateIhis` and `Re_Init`;
 - a switch for the numerical-damping resistors;
 - stock behaviour stays the default.
- **Untracked overlay files:** `tx4_empt.py` and `tx4_case.py`, the custom device kernel and case builder. They are copied from `experiments/parent_empt/overlay/`.

- **Diff hashes.**
 - Patch only: sha256 190b8611...
 - Patch plus overlay, as recorded in every run manifest: a61698cb...

Environment.

- **.venv/xtool-parent:** CPython 3.11.10, with setuptools 72.1.0, xlrd 1.2.0, matplotlib 3.9.2, numba 0.60.0, numpy 2.0.1, pandas 2.2.2 and scipy 1.14.0. Only their dependencies were added; nothing was upgraded.
- **Files:** results/EMT00/environment_manifest.txt, pip_freeze.txt, upstream_provenance.json.
- **Platform:** Windows 11 Pro 10.0.26200, Intel Core Ultra 9 185H, 22 logical CPUs.
- **Solver.** Serial LU; the nxmetis/BBD path was not used or repaired.
- **Reproducibility.** BLAS and numba threads are pinned to 1, which is needed for bit-reproducibility.
- **Phasor references.** Computed in the frozen .venv/tx3-analysis with the canonical device classes, read-only.

EMT00, stock smoke test (results/EMT00/).

- **Run.** IEEE-39 (systemN = 3):
 - network dimension 117;
 - 39 buses, 34 lines, 12 transformers, 10 generators, 19 loads, 2 shunts;
 - $\Delta t = 50 \mu\text{s}$, 261 μs per step.
- **Checks passed:**
 - initialization;
 - determinism (bit-identical);
 - serialization and reload (identical);
 - snapshot and resume;
 - a governor-reference step gives a sensible response (+0.0070 pu P_e).

The exciter step is a no-op in the stock case.
- **Check failed:** stationarity of the no-event run. There is a 0.62 pu voltage jump in the first millisecond.
- **Diagnosis** (tap_diagnostic.json, rule written before the rerun).
 - The stock case carries one off-nominal ratio (6–31, $k = 0.9714$), which upstream never applies.
 - With only that tap applied, the jump is 2.2×10^{-4} pu and the 10-s drift is at the level of the stock power-flow residual.
- **Gate record.** “PASS (tool)”, with the first-written verdict (FAIL) kept in EMT00_gate.json.
- **Disclosure.** This reclassification is a judgment: the defect is in the upstream case and code, not in the installation. It is recorded, not hidden. EMT00 has no TX4 scientific meaning.

5 Official IEEE-39 versus frozen TX4 IEEE-39

The two cases are compared in results/EMT01/network_diff.csv and operating_point_diff.csv; details are in docs/20260911_PAREMT_NETWORK_EQUIVALENCE.md.

	official ParaEMT	frozen TX4
base, frequency, numbering	100 MVA, 60 Hz, 1–39	identical
34 lines (R, X, B); 12 transformer endpoints, R, X; shunts at buses 4, 5	—	identical
off-nominal taps	one (6–31, 0.9714, to-bus side; not applied upstream)	11 taps on bus1 (e.g. 31→6 0.9; 19→33 1.07)
loads	differ at 8 buses (bus 39: 11.04 pu; load 7 Q: 8.4 pu)	ANDES ieee39_full (bus 39: 4.0 pu; load 7 Q: 0.84 pu)
generator ratings / dispatch	1000 MVA each (2000 at bus 39)	S_n 836–1684 MVA, frozen dispatch

Totals. 171 of 195 network rows are identical. The official case is therefore *not* used. A TX4-specific ParaEMT case is built from the frozen data, which is never modified.

6 Equation-equivalent network construction (EMT01)

Construction. ParaEMT’s own numba_InitNet trapezoidal companion models:

- lines as series R–L with $C = b/(2\omega_0)$ at each end;
- transformers as series R–L with an exact ideal tap on bus1;

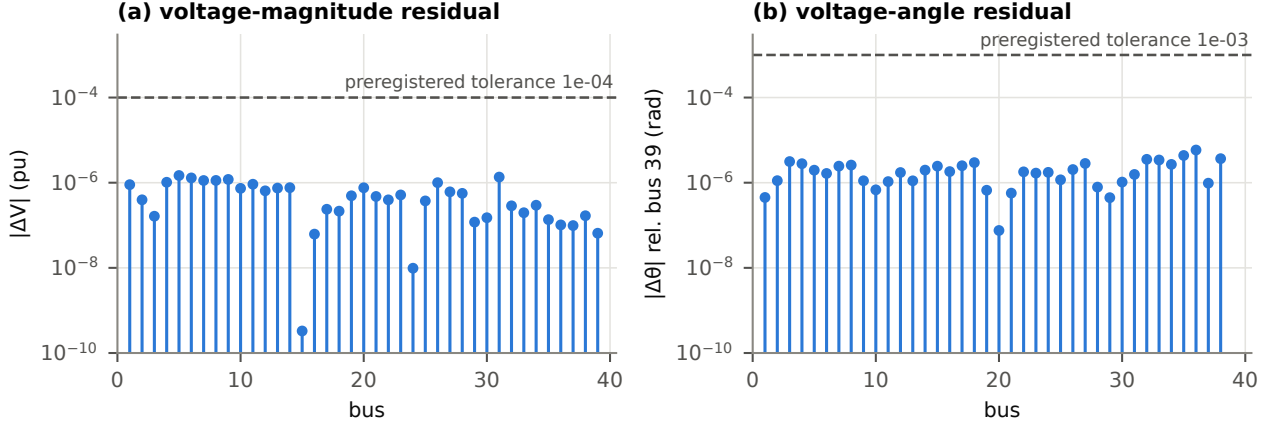


Figure 1: F1 — EMT01 network/model equivalence. Per-bus residuals of the ParaEMT equilibrium against the canonical power flow: base portfolio at P4, 50 μ s, positive-sequence phasors averaged over the last 0.5 s of a 2-s no-event run. Dashed: preregistered tolerances. Bus 39 is the angle reference. Ybus residual 1.7×10^{-16} (tolerance 10^{-8}). Gate G1: PASS.

- shunts as capacitors (`shnt_gb` = $100(g + jb)$).

The numerical damping resistors are switched off: they change the 60-Hz admittance by about 1.4×10^{-3} .

quantity	frozen prediction	EMT measurement	rule	verdict
60-Hz Ybus from stored R/L/C/tap	canonical Ybus	$\max \Delta Y / \max Y = 1.7 \times 10^{-16}$ (25/50/100 μ s)	$\leq 10^{-8}$	PASS
trapezoidal warp at 60 Hz	$(\omega_0 \Delta t)^2 / 12$	$7.4 \times 10^{-6} / 3.0 \times 10^{-5} / 1.2 \times 10^{-4}$	reported	—
equilibrium $ \Delta V $ (P4 base, 2 s no-event, last 0.5 s)	canonical PF	1.48×10^{-6} pu	$\leq 10^{-4}$	PASS
equilibrium $ \Delta \theta $ (rel. bus 39)	canonical PF	5.85×10^{-6} rad	$\leq 10^{-3}$	PASS
device P / Q residual; speed	0	$1.7 \times 10^{-5} / 5.8 \times 10^{-5}$ pu; 8.3×10^{-8} pu	reported	—

Gate G1: PASS. Figure 1 shows the per-bus residuals. Two EMT01 CSV reruns were byte-identical.

Interpretation. The frozen network and its operating point are reproduced by an independent EMT network realization. The only changes are documented convention translations:

- the tap patch fixes an upstream omission;
- damping off;
- the shunt scaling.

This is the only claim-level EMT result of the campaign (I01).

7 Constant-power-load treatment

Outcome A (declared in prereg §3.3): an EMT-compatible fundamental-frequency constant-power load.

- A constant-impedance Norton element $Y_0 = \overline{S_0} / |V_0|^2$.
- A compensating injection $I_{\text{comp}} = -(\overline{S}(t) / V_f - Y_0 V_f)$, where V_f is a 1-ms first-order filter of the extracted phasor.
- The incremental admittance is $Y_0 + (Y_{\text{cp}} - Y_0) / (1 + s\tau_m)$: exact constant power as $s\tau_m \rightarrow 0$, with a relative deviation of 4.4×10^{-3} at 0.7 Hz.

There is no silent substitution; the stock RLC load is not used.

- **Equilibrium with the CP load.** Reproduced in EMT01: PASS.
- **Low-frequency equivalence** (τ_m 0.5 ms vs 1 ms holdout, EMT04 acceptance; gate G2). Not run: UNRESOLVED (BLOCKED).

8 SG equation mapping and unit test (EMT02, gate G3)

Mapping (prereg §3.1). The two-axis machine is `SynchronousMachine` (7 states: $\delta, \omega, e'_q, e'_d, e_{fd}, \text{pss}_w, \text{pss}_l$), with a first-order AVR, a washout-lag PSS, $D = 0$ and no governor.

- Because $x'_d = x'_q$ for all ten machines, the stator $I = (E' - V)/(r_a + jx')$ is realized *exactly* as a synchronous-frame algebraic Norton element:
 - a constant real 3×3 conductance $G_{km} = \frac{2}{3} \text{Re}(a^{m-k}Y)$, with $Y = w/(r_a + jx')$;
 - an EMF current source driven by the predicted states.
- The terminal phasor is $V = \frac{2}{3}(v_a + av_b + a^2v_c)e^{-j\omega_0 t}$.

Test system (prereg §7 and amendment A1).

- **Line.** The device at T, then an EMT R–L line ($R = 0, X = 0.05$ pu), then INF.
- **Infinite bus.** Thevenin $1.0\angle 0$ behind $j10^{-4}$ pu, solidly grounded through a zero-sequence-only conductance.
- **Operating point.** $S = 4 + j1$ pu, bus-30 data, P4 settings.
- **Disturbance.** $P_m + 2\%$ for 0.2 s at 1 s; 10-s run at 50 μ s.
- **Reference.** The canonical class integrated by Radau (rtol 10^{-10}) with a quasi-static network.

quantity	reference	EMT	rule	verdict
equilibrium $\Delta P, \Delta Q$	0	$-8.3 \times 10^{-5}, +2.3 \times 10^{-5}$ pu	$\leq 10^{-4}$	pass
$\delta, \omega, e'_q, e'_d, \text{pss}_w, \text{pss}_l$: error/excursion	0	0.0032, 0.0026, 0.0087, 0.0018, 0.0013, 0.0016	≤ 0.02	pass
e_{fd} : error/excursion	0	0.0226	≤ 0.02	fail
ringdown α, f (matrix pencil on ω)	$-0.11837 \text{ s}^{-1},$ 1.17711 Hz	$-0.11868 \text{ s}^{-1}, 1.17723 \text{ Hz}$	$ \Delta\alpha , \Delta f \leq$ 0.005	pass

Gate G3: FAIL. No threshold was changed.

Diagnostics (post hoc; they do not lift G3). Source: `results/EMT02/EMT02_diagnostics.json`; see Figure 2.

- **D1.** The T–INF line realized as an algebraic synchronous-frame Norton element, so the EMT network is algebraic. Worst state ratio 1.9×10^{-4} : the machine transcription is exact to integration accuracy.
- **D2.** The preregistered EMT line at 25 μ s. e_{fd} ratio 0.0228: the excess does not shrink with Δt , so it is not a discretization error.
- **D3.** The preregistered EMT line against a phasor reference that adds the line's $(X/\omega_0) dI/dt$. Worst ratio 1.4×10^{-3} : the excess is the electromagnetic dynamics of the line.

Interpretation.

- The error oscillates at the 1.18-Hz electromechanical frequency, with relative size $f/f_0 = 1.96\%$. This is the known first-order dynamic-phasor correction that the quasi-static phasor model omits.
- The preregistered 2% rule was therefore marginal for a comparison that deliberately crosses modelling domains.
- Because the failure is attributed post hoc, the gate stays FAILED.

9 GFL equation mapping and unit test (EMT03, gate G4)

Mapping. The equation-by-equation map is in `docs/20260911_PAREMT_GFL_EQUATION_MAP.md`:

- all 11 states ($\theta, x_{pll}, p_f, q_f, x_p, x_q, i_d, i_q, x_{id}, x_{iq}, x_v$);
- the algebraic quantities, parameters and initialization, identical to `GridFollowingConverter` (GFL spec §4.2);
- the average-value interface: dq voltages from the extracted phasor, output $I = w(i_d + ji_q)e^{j\theta}$ injected as an ideal three-phase current source.

No stock converter is used, and no limits, DC link, PWM or protection are added.

Test. The same system as the SG test, with $g \in \{0, 0.03625, 0.25\}$ and three disturbances at 1 s:

- (a) $p_{\text{ref}} + 1\%$;
- (b) $|E| - 1\%$;
- (c) E phase $+0.02$ rad.

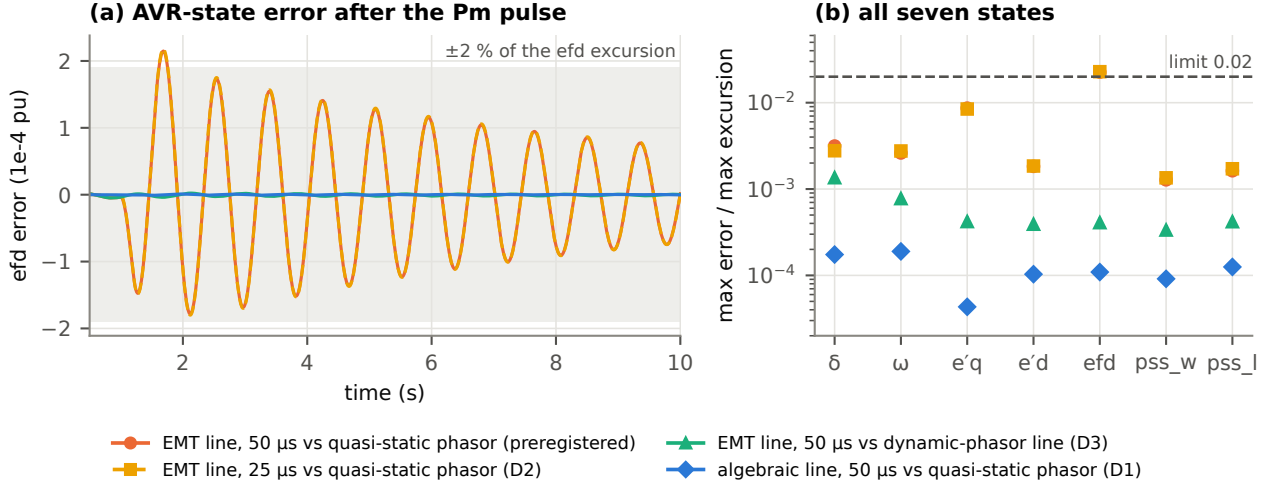


Figure 2: FD1 — EMT02 SG unit test (diagnostic figure, not an F-series result). (a) e_{fd} error after the P_m pulse; grey band: $\pm 2\%$ of the e_{fd} excursion. (b) Maximum error over maximum excursion for all seven states. Orange: the preregistered test (G3 FAIL on e_{fd}). Yellow: 25 μ s. Aqua: against a reference that includes the line's dI/dt . Blue: the line realized algebraically.

Each run is 5 s at 50 μ s.

quantity	rule	EMT measurement	verdict
every state, 9 cases	$\text{error} \leq 0.02 \times \text{excursion}$	3/99 state checks pass; $g = 0$ non-finite at $t = 0.76$ s (before the disturbance); $g > 0$ worst-state ratios 3.2–1103	fail
equilibrium P/Q	$\leq 10^{-4}$	$g = 0$: non-finite; $g = 0.03625$: $\Delta P = 2.5 \times 10^{-3}$, $\Delta Q = -0.31$; $g = 0.25$: $\Delta P = -0.34$, $\Delta Q = 0.35$ pu	fail
PLL steady state	$ \theta - \angle V_T \leq 10^{-3}$	0.19–0.33 rad ($g > 0$); non-finite ($g = 0$)	fail
$g = 0$: $q_f \rightarrow q_{\text{ref}}$	$\leq 10^{-3}$	non-finite	fail

Gate G4: FAIL.

Diagnostics. See Figure 3.

- **D4, mechanism.** With no disturbance, the step-to-step alternation of $|V_T|$ grows:
 - at $g = 0$: exponentially at 26.6 s^{-1} (factor 1.0013 per step) until the run fails;
 - at $g = 0.25$: at 445 s^{-1} , saturating at about 0.11 pu by 0.1 s.
 - Reading: an ideal current source feeding a lossless trapezoidal inductor has an undamped Nyquist mode, and the explicitly coupled converter loop amplifies it.
- **D1, transcription.** With the line algebraic, all nine cases are finite and every state is within 0.78% of its excursion. The transcription is correct; the failure is in the EMT interface.
- **D5, IEEE-39.** No-event 30-s runs of the preregistered realization: P4 base, P4 H4, and H4 at G_S .
 - The start-up alternation at the four GFL buses (5×10^{-6} pu) decays to 2×10^{-8} (P4 H4) and 7×10^{-9} pu (G_S H4) over 1–10 s. No step-to-step instability develops.
 - In P4 H4 over 10–30 s, the consecutive-sample difference rises to 2.4×10^{-7} pu, together with a slow $|V|$ drift of 6.8×10^{-5} pu. This smooth growth was *not analysed*.
 - No estimator was applied and no phasor prediction was compared. **D5 does not lift G4.**

10 EMT initialization

Procedure (prereg §4).

- The network phasors come from the canonical power flow.
- The device states come from the phasor `initialize` formulas.
- ParaEMT sets the branch histories from the phasor steady state.
- The load filter starts at $V_f = V_0$.

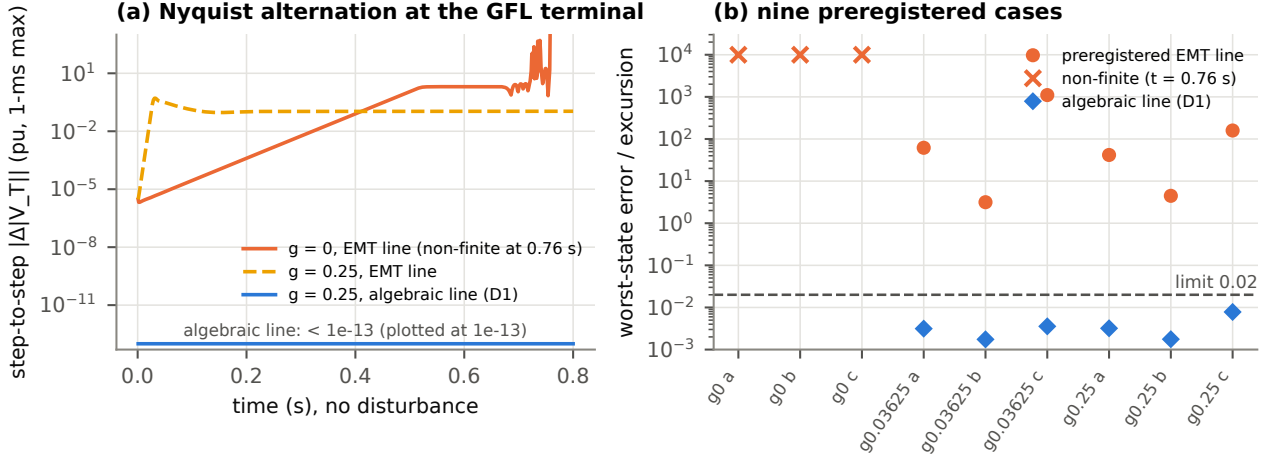


Figure 3: FD2 — EMT03 GFL unit test (diagnostic figure). (a) Step-to-step alternation of the terminal $|V_T|$ (1-ms maxima) with no disturbance: the Nyquist mode grows with the preregistered EMT line and is absent with the line realized algebraically. (b) Worst-state error over excursion for the nine preregistered cases: preregistered EMT line (orange; $\times$ = non-finite) versus the algebraic-line diagnostic D1 (blue). Gate G4: FAIL.

Verified.

- In EMT01, the P4 base drifts by at most 1.5×10^{-6} pu.
- In the unit tests, the equilibrium P/Q residual is 8.3×10^{-5} pu (SG).
- The GFL equilibrium was destroyed by the chatter of Section 9.

11 Numerical convergence (EMT04, gate G5)

Frozen rule. The verdicts of base, 30+33+35, H4 (P4) and H4 (G_S) must be invariant between 50 and 25 μ s, with $|\Delta f| \leq 0.01$ Hz and $|\Delta \alpha| \leq 0.01$ s $^{-1}$.

Measurement. Not run: UNRESOLVED (BLOCKED) by G3/G4.

Interpretation.

- The only Δt evidence is the SG unit test: the e_{fd} ratio is 0.0226 at 50 μ s and 0.0228 at 25 μ s (D2). That is consistent with convergence but is not an EMT04 result.
- The 60-Hz warp is 3.0×10^{-5} at 50 μ s.

12 Common disturbance and modal estimator

Frozen (prereg §5), not exercised on any TX4 portfolio case.

- **Disturbance.** +2% of the bus-20 active load (0.136 pu) on [1.0, 1.2) s.
- **Signals.** Relative machine speeds, GFL frequency deviations, and bus-angle differences and $|V|$ at the ten generator buses.
- **Estimator A.** Multi-channel matrix pencil at 10 Hz.
- **Estimator B.** Hilbert envelope of the dominant band-passed channel at 50 Hz.
- **Band and window.** 0.2–1.2 Hz, window [2.2 s, T].
- **Classification.** STABLE / UNSTABLE if both estimators are resolved and beyond $\pm \varepsilon = 0.002$ s $^{-1}$; UNRESOLVED otherwise.
- **Implementation.** experiments/parent_empt/empt_estimator.py, used here only for the SG ringdown.

13 EMT05: P4 portfolio trap

All experiments in Sections 13–22 were preregistered with frozen phasor predictions (results/EMT_PRED/phasor_predictions.csv, commit c2947bd8). None was run, because gate A (MODEL) failed at G3 and G4.

For each experiment:

- EMT measurement: none;
- verdict: UNRESOLVED (BLOCKED);

- interpretation: no EMT evidence either way.

Prediction.

- The 15 proper subsets of $\{30, 33, 35, 37\}$ (including $\emptyset$) are stable, with in-band α from -0.214 to -0.144s^{-1} at $0.64\text{--}0.92\text{Hz}$.
- H4 is unstable: $\alpha = +0.1270\text{s}^{-1}$ at 0.622Hz .

Rule.

- All 15 proper subsets STABLE and H4 UNSTABLE ($\kappa_{\text{EMT}} = 4$).
- Secondary: $\text{MAE}(\alpha) \leq 0.03\text{s}^{-1}$ and $|\Delta f| \leq 0.05\text{Hz}$.
- Amplitude checks at 1 % and 4 %.

14 EMT06: contextual intervention sign reversal

Prediction. Each of 30, 33, 35 and 37 is stabilizing alone and destabilizing when added last (the signs of 8 effects).

Rule. All 8 EMT signs equal the phasor signs.

15 EMT07–08 (and EMT09, EMT10): g-only controller boundary

Prediction. At H4, $k = 1.425$, $\alpha(g) = +0.0138, +0.0036, +0.0012, -0.0011, -0.0076, -0.0174\text{s}^{-1}$ at $g = 0.18, 0.20, 0.205, 0.21, 0.225, 0.25$; $g^* = 0.20768$.

Rule. A resolved UNSTABLE point below and a STABLE point above g_{EMT}^* . Target: $|g_{\text{EMT}}^* - 0.20768| \leq 0.02$.

Also blocked.

- **EMT09** (P_∞ H4): prediction $\alpha = -0.155\text{s}^{-1}$; rule STABLE.
- **EMT10** (Newton design):
 - prediction: the first iterate $g = 0.08806$ has $\alpha = +0.0739$; g^* has $\alpha = 0$; $g = 0.25$ has $\alpha = -0.0174$;
 - rule: α_{EMT} monotone along the sequence, $g = 0.25$ STABLE, and $|g_{\text{EMT}}^* - g^*| \leq 0.02$.

16 EMT11: k boundary

Prediction. At $g = 0.03625$, $\alpha(k) = -0.0809, -0.0337, -0.0061, +0.0069, +0.0313, +0.0539$ at $k = 1.25\text{--}1.35$; $k^* = 1.3046267$.

Rule. $|k_{\text{EMT}}^* - 1.3046267| \leq 0.02$, with a clear STABLE $\rightarrow$ UNSTABLE change.

17 EMT12: line reinforcement

Prediction. Exact phasor finite changes $\Delta\alpha$ for 12 frozen holdout lines $\times 1.5$ at P4 H4:

- from -0.0371 (L42), -0.0340 (L35) and -0.0242 (L26);
- to $+0.0013$ (L17) and $+0.0042$ (L31).

Rule. Sign agreement $\geq 10/12$ and Spearman ≥ 0.80 .

Not claimed. Transfer across converter models.

18 EMT13: condenser

Prediction. At P4 H4, the damped condenser at $2.0/2.5/3.0/5.0\%$ gives in-band $\alpha = +0.0264/ - 0.0010/ - 0.0295/ - 0.1569\text{s}^{-1}$.

Rule. 2 % UNSTABLE and 5 % STABLE, in band.

19 EMT14: governors

Prediction.

- Governed H4 at P4: $\alpha = -0.0745 \text{ s}^{-1}$ (stable).
- Governed H4 at $g = 0.020$: $\alpha = +0.0691 \text{ s}^{-1}$ (unstable, descriptive).

Rule. The governed P4 H4 is STABLE.

20 EMT15: nonlinear recovery near the subcritical Hopf

Prediction. H4 at $g = 0.22$ ($\alpha = -0.0055$) and $g = 0.30$ (-0.0334).

Protocol. Pulse amplitudes of 2–80 % with RETURNS / LEAVES / FAILS / NUMERICAL labels.

Rule. Characterize the outcome, whether positive or negative.

21 EMT16: robustness holdout

Prediction. From the frozen PCV05 draws (seed 20260921; 74 cases):

- A: the witness is H4;
- B: the witness is $\{30, 33, 35\}$;
- C: H4 is stable.

Rule. Each set passes in ≥ 80 % of its draws.

22 EMT17: optional Kundur zero-frequency check

Not run: it is optional, and the campaign is blocked. T08 rests on its proof and on the existing phasor holdout: 28/29 real-count changes under the preregistered rule, i.e. 28/28 crossings through the origin, with 0/445 false positives.

23 Claim-to-EMT matrix

File: `results/20260911_EMT_CLAIM_MATRIX.csv`, generated by `experiments/parent_empt/EMT_claim_matrix.py` (regenerated in V2-0 from the final ledger).

Rows (43):

- the final canonical ledger `results/20260911_FINAL_VALIDATION_MATRIX.csv` (V01–V30);
- the 13 older ids that no V row cross-references (T04, T06, T08, C08, B04, B06, B07, B10, N01, N05–N08), from `results/20260911_CLAIM_MATRIX.csv`.

Both ledgers are read only. The column `independent_phasor_reproduction` is kept separate from `EMT_result`.

EMT label	claims
THEOREM	V01, V02, V08, V22, V26, T04, T06, T08 (8)
NOT APPLICABLE TO EMT	V05, V09, V12, V23, V24, V25, V27, V28, C08, N01, N05–N08 (14)
EMT QUANTITATIVELY REPRODUCED	V19 (1)
EMT UNRESOLVED	V03, V04, V06, V07, V10, V11, V13–V18, V20 (G3 FAIL), V21 (G4 FAIL), V29, V30, B04, B06, B07, B10 (20; BLOCKED except V18 and B06, which were not EMT targets)
EMT CORROBORATED / REFUTED	none

24 Negative and unresolved results

Negative (preregistered gates failed).

1. **G3.** The SG e_{fd} trajectory error is 2.26 % against a 2 % limit.
2. **G4.** The GFL unit test is numerically unstable with the preregistered EMT line: 3/9 runs non-finite, 6/9 grossly wrong.
3. **EMT00.** The stock no-event stationarity check failed as first written (upstream never applies the transformer tap).

Unresolved. EMT04–EMT18 are BLOCKED. Every portfolio, policy, design, remediation, nonlinear and robustness question remains unanswered by EMT.

Not done.

- No retuning.
- No threshold change.
- No stock GFL or machine substitution.
- No constant-power-load substitution.
- No IEEE-39 portfolio comparison.

25 Limitations

- **Diagnostics are post hoc.** D1–D5 were designed after the gate failures. They explain the failures; they do not replace the gates.
- **The unit-test system is part of the preregistration.** A lossless line to a stiff source is a demanding topology for an ideal current source. D5 shows the IEEE-39 GFL buses behave differently, but the test was chosen in advance and failed.
- **The GFL interface is a design choice.** It is an average-value ideal current source with explicit predictor–corrector coupling. Other interfaces were not preregistered and were not tried as a gate:
 - a converter voltage behind an EMT filter branch;
 - Norton-compensated injection;
 - ParaEMT’s standard numerical damping.
- **A domain-crossing comparison.** The 2 % trajectory rule compares a quasi-static phasor reference with an EMT network. The network’s own electromagnetic dynamics contribute about f/f_0 of the excursion.
- **Scope of EMT01.** It covers the base portfolio at P4 only; the GFL equilibrium in IEEE-39 was not gated.

26 Manuscript implications

No wording changes.

- The manuscript already states “nonlinear phasor-domain simulation (not EMT)” (`docs/FINAL_IAS_SAFE_CLAIMS.md`). That wording remains correct and required.
- The manuscript must *not* cite this campaign as EMT validation of any portfolio, policy or design result.
- The custom-GFL results were reproduced in a second phasor-domain tool (ANDES, same equations; V21, Phase 8). They were *not* reproduced in EMT. (*Corrected in V2-0; the first version wrongly said “NOT YET TESTED / single-implementation”.*)

Optional sentences (limitations / reproducibility):

- “*The IEEE-39 network and operating point were independently realized in the ParaEMT EMT solver (60-Hz admittance residual 2×10^{-16} , equilibrium residual 2×10^{-6} pu).*”
- “*An equation-equivalent EMT realization of the converter did not pass its preregistered unit test; EMT corroboration of the portfolio results is not claimed.*”

27 Reproducibility instructions

Paths are relative to `reports/poster/ias2026/research/`. EMT runs use `.venv/xtool-parent`; phasor references use `.venv/tx3-analysis` (read-only).

1. **Clone and patch.** Clone ParaEMT_public at d79d735a into `external/ParaEMT_upstream` and `external/ParaEMT_tx4`; run `tx4_patch_parent.py` and `EMT_setup_tx4.py`.
2. **EMT00.** `EMT00_provenance.py`, `EMT00_stock_smoke.py`, `EMT00_tap_diagnostic.py`, `EMT00_gate.py`.

3. **Frozen predictions** (already committed): `EMT_phasor_predictions.py`, `EMT_export_operating_point.py`.
4. **EMT01**. `EMT01_network_equivalence.py`.
5. **Unit-test references** (tx3-analysis): `EMT02_03_phasor_refs.py`, `EMT02_03_dynphasor_ref.py`.
6. **Unit tests and diagnostics**: `EMT02_03_unit_tests.py`, `EMT02_03_diagnostics.py`, `EMT03_D5_ieee39_chatter.py`. Then `EMT02_03_manifest.py` re-executes all three and writes the manifest.
7. **Claim matrix and figures**. `EMT_claim_matrix.py`, `EMT_figures.py`.
8. **Report**. Build this file with `pdflatex`, twice.

Determinism.

- Every summary CSV/JSON of EMT01–EMT03 and the diagnostics was regenerated twice and is byte-identical. Wall-clock fields sit only in the manifests.
- The phasor references are bit-identical after pinning BLAS to one thread. Before pinning they differed by up to 7×10^{-12} .
- The claim matrix and the figure sources are byte-identical on rerun.

Manifests.

- EMT01: `results/EMT01/EMT01_summary.json`.
- Unit tests and diagnostics: `results/EMT02/EMT02_03_manifest.json`.
- Raw EMT01 series: `external/parent_runs/raw/`. Unit-test trajectories: `results/EMT02/emt_traj.npz`, `results/EMT03/emt_traj.npz`.

Figures.

- F1 is Figure 1; FD1 and FD2 are diagnostic figures.
- F2–F8 need EMT04–EMT16 and are **not produced (BLOCKED)**.
- No cumulant figure.

Answers to the final questions

1. **Is the frozen TX4 IEEE-39 faithfully reproduced in ParaEMT?** Yes, for the network and the operating point:
 - Ybus residual 1.7×10^{-16} ;
 - equilibrium $|\Delta V|$ 1.5×10^{-6} pu and $|\Delta \theta|$ 5.9×10^{-6} rad.
 This required a documented tap patch. The official ParaEMT IEEE-39 is a different case.
2. **Is the custom GFL equation-equivalent in EMT at fundamental frequency?** Not established.
 - The equations are transcribed identically, and match within 0.78 % with an algebraic line (diagnostic).
 - But the preregistered unit test failed (G4): the ideal-current-source interface is unstable with the pre-registered EMT line.
3. **Is the two-axis SG equation-equivalent?** Not established by the gate.
 - G3 failed on e_{fd} (2.26 % against 2 %); the other states pass and the ringdown agrees to $3 \times 10^{-4} \text{ s}^{-1}$.
 - The diagnostics suggest equivalence, but that is a post hoc reading.
4. **Proper subsets stable and H4 unstable?** Not tested (BLOCKED).
5. κ_{EMT} at P4? Unknown (BLOCKED).
6. α_{EMT} , f_{EMT} for the 16 subsets? Not measured (BLOCKED); predictions in Section 13.
7. **Contextual sign reversal?** Not tested (BLOCKED).
8. **g-only transition?** Not tested (BLOCKED).
9. g_{EMT}^* vs 0.20768? Not measured (BLOCKED).
10. k boundary? Not tested (BLOCKED).
11. **Port-guided retuning?** Not tested (BLOCKED).
12. **Line sensitivities vs finite EMT reinforcement?** Not tested (BLOCKED).
13. **Condenser remediation?** Not tested (BLOCKED).
14. **Governor stabilizes the flagship?** Not tested (BLOCKED).
15. **Finite recovery threshold near the subcritical Hopf?** Not tested (BLOCKED); no EMT evidence either way.
16. **Robustness holdout?** Not tested (BLOCKED).
17. **Which claims are not EMT-testable?**
 - T01–T07 and C01–C06, which remain proofs.
 - T08, whose optional EMT17 was not run.
 - The identities behind T09 and T10: only the observable consequences of T10 are EMT-testable.
 - C07–C11, phasor-port constructions with no EMT observable.

18. **Did any manuscript statement become too strong?**
 - None because of EMT: no portfolio-level EMT result exists.
 - The “phasor-domain, not EMT” wording must stay, and the campaign must not be cited as EMT validation.
 - The omitted line dI/dt is now measured: deviations of about f/f_0 (2% at 1.2 Hz), within the declared “no claim above a few Hz” scope.
19. **New model-fidelity limitations?** Yes, three:
 - (i) The average-value GFL, as an ideal current source with explicit coupling, is numerically fragile in EMT (Nyquist chatter) when its terminal sees only lossless inductance to a stiff source.
 - (ii) The quasi-static network omits the line’s dI/dt (about f/f_0 on machine trajectories).
 - (iii) Upstream ParaEMT ignores transformer ratios, and its IEEE-39 case is not TX4.
20. **Strongest defensible statement:**
 - **Mathematical proof:** the theorems stand.
 - **Python phasor model:** the frozen phasor results stand.
 - **ANDES:** it independently reproduces, with the same equations, the network, the synchronous dynamics and branch sensitivities (V19, V20), and the custom-GFL portfolio results: P4 trap 32/32, H and κ equal, P4 vs G_S (V21, Phase 8).
 - **Nonlinear phasor TDS:** it confirms the small-signal verdicts (B09).
 - **ParaEMT:** it independently reproduces the network and operating point (I01), but provides *no* corroboration of the portfolio, policy, design, remediation or nonlinear claims. Those claims remain phasor-domain results, reproduced in two phasor tools (V21) but not in EMT. (*Corrected in V2-0.*)

Appendix A. Proposed preregistration v2 (for the author’s decision; not executed)

Superseded (V2-0). The author approved a different V2. It separates device transcription from EMT embedding (G3a/G3b, G4a/G4b) and preregisters a single voltage-source-behind- R_f-L_f GFL interface. Numerical damping and Norton compensation are excluded as primary solutions. This appendix is kept only as a record and is not executed.

Any v2 is designed *after* seeing the v1 failures and diagnostics. It must say so, and it must add blind elements so that it is not a forced match.

V2-1. Device gates on an algebraic-network unit test.

- The line is realized as a synchronous-frame algebraic Norton element, so the test isolates the device transcription. The 2% rule and the equilibrium, ringdown and PLL rules are unchanged.
- Blind holdout, fixed in v2 before any run: a second operating point and device (e.g. bus-35 data, $S = 2 + j0.5$) and a second disturbance set.
- **Caveat.** v1’s D1 already passed on the v1 operating point, so only the holdout is blind.

V2-2. Network electromagnetic effect as a reported quantity, not a gate.

The EMT-line unit test is compared against the dynamic-phasor reference.

V2-3. One preregistered GFL EMT interface, chosen before any IEEE-39 run.

Candidates:

- (a) the converter voltage behind the filter reactor realized as an EMT R–L branch; the θ' cross-coupling difference must be bounded;
- (b) the ideal current source with ParaEMT’s standard numerical damping on; the 1.4×10^{-3} admittance change is declared and checked by a holdout;
- (c) Norton-compensated injection.

Acceptance: the preregistered EMT-line GFL unit test must be finite and stable in all nine cases, and V2-1 must pass.

V2-4. Unchanged.

EMT04–EMT16 as in v1, with unchanged thresholds, disturbance, estimator and frozen predictions.

V2-5. Declared history.

v1 failed G3 and G4; that record stays in the ledger regardless of v2.

What EMT did and did not validate

The table separates **theorem validity** (mathematical status) from **model-fidelity corroboration** (phasor, independent and EMT columns). *Rebuilt in V2-0 from the final canonical ledger (V-ids, with old ids in brackets).*

Theory / claim	Mathem- atical status	Phasor evidence	Independent phasor evidence	EMT evidence	Final status
V01 transverse quotient [T01]	proved	S1 (4.6×10^{-10})	ANDES structural pair 32/32	not applicable (THEOREM)	PROVED + REPRODUCED
V02, V26, T04, T06 (antichain / any-order planning, NP, Metzler, complex)	proved	used throughout	—	not applicable (THEOREM)	PROVED
V08 factorization [T07, T09]; V22 cumulant theorems [C01–C06]	proved	PCV03, FC18; CC02	—	not applicable (THEOREM)	PROVED (+ benchmark-specific boundary statement, V08)
T08 zero-frequency port	proved	Kundur holdout 28/29 (28/28 origin crossings), 0/445 FP	—	EMT17 not run	PROVED
V03 P4 trap, $\kappa = 4$ [B02]	—	PCV02; G2 TDS	ANDES Phase 8: 32/32, H , κ equal	BLOCKED (EMT05/06)	VALIDATED (model-specific witness)
V04 lower-order screening approves H4	follows from V03	PCV02/03	via V03	BLOCKED (EMT05)	VALIDATED
V06, V07 policy-dependent incompatibility [B01]	structure proved	clean g-only counterfactual; F7	ANDES Phase 8: P4 vs G_S	BLOCKED (EMT07–08/10/11/16)	VALIDATED
V10 port derivative, Newton [T10]	proved	PCV04 (g , k Newton)	ANDES R3 11/12, ρ 1.00	BLOCKED (EMT08/10/11/12)	PROVED + VALIDATED
V11 line ranking	—	ρ 0.937; finite sign 9/12	line sensitivities equation- equivalent in ANDES	BLOCKED (EMT12)	VALIDATED (local ranking)
V13 remediation, V14 governors [B03, B05]	—	PCV04; FC03; G2	ANDES Phase 8 (g side); governor direction	BLOCKED (EMT08/10/11/13/14)	VALIDATED (model-specific); BENCHMARK- SPECIFIC
V15–V18 robustness	—	PCV05	—	BLOCKED (EMT16) / not targeted (V18)	R1 PASS 3/4; R2 NOT ROBUST; R3 descriptive; R4 mixed
V30, B07, B10 nonlinear scope [B08, B09]	normal form	G2 TDS 32/32; FC05–07; FC13	—	BLOCKED (EMT15, EMT05)	NUMERICALLY VALIDATED
V19 network and operating point [I01]	—	reference	ANDES 10^{-13} , pandapower 3×10^{-14} after tap fix	EMT QUANTITA- TIVELY REPRODUCED (Ybus 2×10^{-16} , V 1.5×10^{-6})	CONDITIONAL (pandapower translation; memo A)
V20 SG dynamics and sensitivities [I02, I03]	—	internal	ANDES R2/R3 12/12; E1 12/12	G3 FAIL (e_{fd} 2.26%) → EMT UNRESOLVED; EMT12 BLOCKED	VALIDATED
V21 custom-GFL results in a second simulator [I04]	—	project DAE	ANDES Phase 8, GATE 5 PASS (same equations)	G4 FAIL → EMT UNRESOLVED	VALIDATED as a reproduction of computation; EMT reproduction not achieved
V09, V23, V24, V25, C08 (cumulants, closure diagnostics)	V22	CC campaign; PCV02	—	not applicable	BENCHMARK- SPECIFIC / DIAGNOSTIC / REFUTED
V05 F10 recount; V12 finite-step nonlinearity B04, B06	—	PCV02; PCV04	—	not applicable	VALIDATED; NEGATIVE RESULT
	—	FC04; FC10	—	BLOCKED / not targeted	BENCHMARK- SPECIFIC
V27, V28, V29, N01, N05–N08 negatives	—	refuted	—	not applicable / BLOCKED (V29)	REFUTED (kept)